NETWORKWALKS

PENETRATION TESTING REPORT

FOOTPRINTING, RECONNAISSANCE & NETWORK SCANNING

REPORT DETAIL	INFORMATION
Prepared by	Debela Beyena
Program / Batch	Cybersecurity Program – B084, Networkwalks
Project	W2-PM-FINAL | Week 2 Practical Modules
Modules completed	W2-PM1 – Multiple Kali Tools; W2-PM5 – Zenmap Scanning
Report date	12 November 2026

Prepared for educational and authorized cybersecurity training purposes.

1. Liability Disclaimer

All activities described in this report were stated to have been performed only against systems for which written permission had been obtained or systems owned by the author. The exercises are intended for education and research. Unauthorized access or scanning may violate applicable laws and organizational policies. The author is responsible for ensuring that all testing remains within the approved scope.

2. Executive Summary

This report documents two Week 2 practical activities completed as part of the Networkwalks cybersecurity internship: (1) public-domain footprinting of networkwalks.com using Kali Linux tools, and (2) host discovery on the author's own local network using Zenmap (Nmap GUI). The activities demonstrate how publicly available domain information and network responses can be collected and documented before any vulnerability validation or exploitation.

The footprinting exercise reported domain-registration details, web-technology indicators, DNS resolution, HTTP response information, WAF identification, and DNS-record enumeration. The network-scanning exercise used a Ping Scan against the 10.0.0.0/24 subnet. The supplied Zenmap screenshot visibly shows three responding hosts: 10.0.0.1, 10.0.0.2, and 10.0.0.4.

3. Objectives

- Collect publicly available registration and DNS information for the specified domain.
- Identify website technology indicators and inspect basic HTTP response information.
- Identify whether a Web Application Firewall (WAF) indicator is present.
- Discover responding hosts on the authorized local subnet using Zenmap.
- Document the observed results and supporting evidence without overstating security conclusions.

4. Scope and Methodology

4.1 Domain Footprinting

Target: networkwalks.com. The source report states that written permission was secured. The work was limited to passive or low-impact information gathering and service-response inspection as described in the practical.

4.2 Local Network Scanning

Target: the author's own local LAN, using the subnet 10.0.0.0/24 shown in the Zenmap evidence. The scan profile was Ping Scan, intended to identify responding hosts rather than perform an exhaustive service or vulnerability assessment.

5. Tools and Their Purpose

TOOL	PURPOSE
Kali Linux	Operating-system environment used for domain footprinting.
WHOIS	Retrieve publicly available domain registration and

	registrar information.
WhatWeb	Identify web technologies and application fingerprints exposed by the website.
nslookup	Resolve the domain name through DNS.
curl -I	Inspect HTTP response headers returned by the web server.
Wafw00f	Check for indicators of a Web Application Firewall.
DNSRecon	Enumerate DNS records and related infrastructure information.
Zenmap / Nmap	Discover responding hosts on the authorized local subnet.
Windows ipconfig	Identify the local computer's IP configuration and subnet.

6. Activities Performed and Findings

6.1 Footprinting and Reconnaissance

6.1.1 WHOIS

WHOIS was used to retrieve publicly available registration information for networkwalks.com, including registrar and name-server details. This information can help establish the domain's registration and DNS context.

6.1.2 WhatWeb

WhatWeb was used to fingerprint the website. The original report records WordPress 7.0.4 and WP Download Manager 3.3.58 among the identified technologies. Technology/version disclosure can help a tester identify components that may warrant patch and configuration review; it does not, by itself, prove that a vulnerability exists.

6.1.3 DNS Resolution – nslookup

DNS resolution returned the IPv4 address 192.232.216.135 for networkwalks.com. DNS resolution identifies an address associated with the queried domain at the time of testing; it should not be interpreted as proof that the address is a dedicated or permanent host.

6.1.4 HTTP Response Inspection – curl -I

The command was used to inspect HTTP response headers. The evidence shows an HTTP 200 response and headers that include server/application-related information. The report also records the presence of the WordPress REST API path /wp-json/.

6.1.5 WAF Identification – Wafw00f

The tool identified the website as being behind a ModSecurity (SpiderLabs) WAF. This is an observed technology indicator, not a test of the WAF's effectiveness or coverage.

6.1.6 DNS Enumeration – DNSRecon

DNSRecon was used to enumerate DNS information, including name-server, mail-server, TXT/SPF, and service-related records. The source report describes these categories of records but does not provide a complete record-by-record output in the extracted report text.

6.2 Network Scanning with Zenmap

The local network configuration was first checked using the Windows ipconfig command. The subnet shown in the Zenmap evidence is 10.0.0.0/24. A Ping Scan was then performed using the Nmap command displayed in Zenmap:

nmap -sn 10.0.0.0/24

The supplied screenshot shows the following responding hosts:

IP ADDRESS	EVIDENCE-BASED OBSERVATION
10.0.0.1	Host reported up; a MAC address is displayed in the screenshot.
10.0.0.2	Host reported up; no MAC address is clearly visible in the supplied screenshot.
10.0.0.4	Host reported up; a MAC address is displayed in the screenshot.

7. Limitations and Considerations

- The findings are limited to the outputs and screenshots retained in the source report.
- The DNSRecon section does not include a complete record-by-record output in the extracted report text.
- A Ping Scan identifies hosts that respond to the scan; it does not establish that non-responding hosts are absent or that discovered hosts expose particular services.
- Technology fingerprints and WAF identification are indicators only and do not establish the presence or absence of exploitable vulnerabilities.
- The observations represent the environment at the time of testing and may change.

8. Conclusion

The Week 2 practical exercises provided hands-on experience with two foundational penetration-testing activities: footprinting and network discovery. Using Kali Linux tools, the domain reconnaissance exercise gathered registration, technology, DNS, HTTP, and WAF-related information about networkwalks.com. Using Zenmap, the local-network exercise identified responding hosts on the 10.0.0.0/24 subnet.

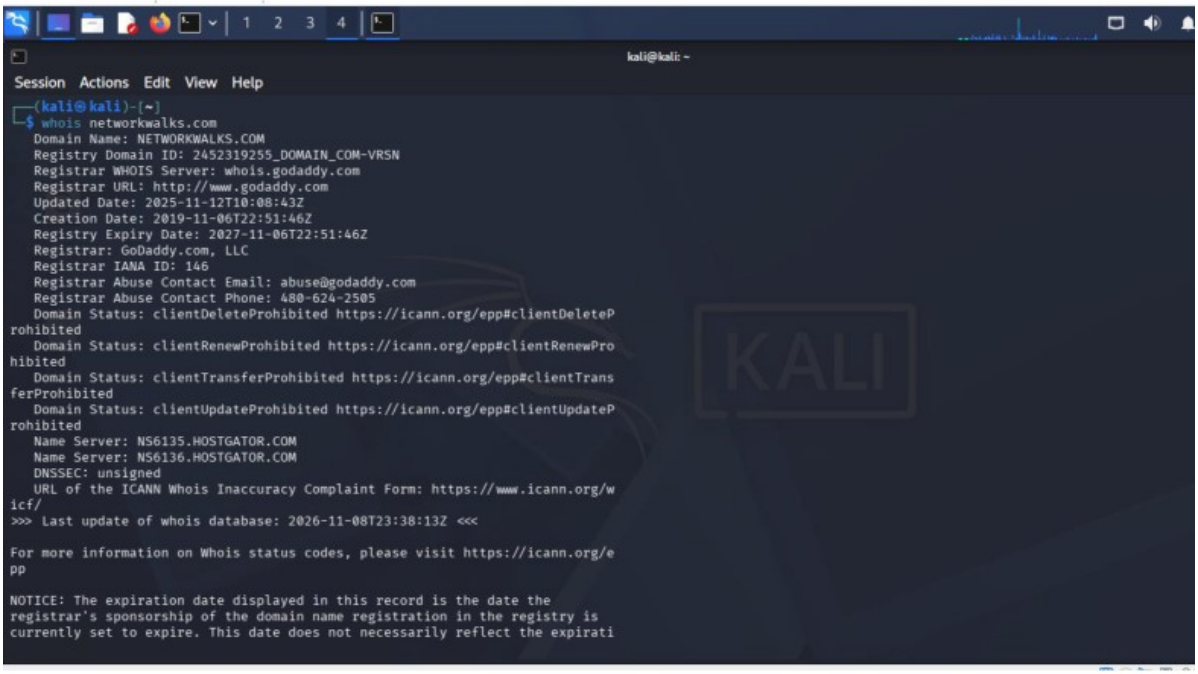
The work demonstrated how publicly available information and basic network responses can help build an initial picture of an environment. It also reinforced the importance of documenting commands, observed results, limitations, and security implications accurately. The findings in this report are preliminary observations rather than proof of exploitable weaknesses. Any further testing should be performed only with explicit authorization and within a clearly defined scope.

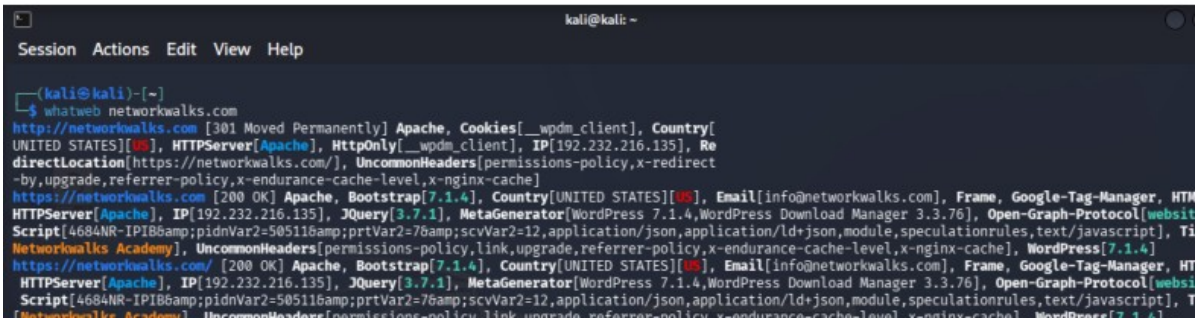
9. Evidence Collected

The following screenshots were retained from the original report as supporting evidence. They should be checked against the final command outputs before formal submission.

Evidence 1 – WHOIS and WhatWeb

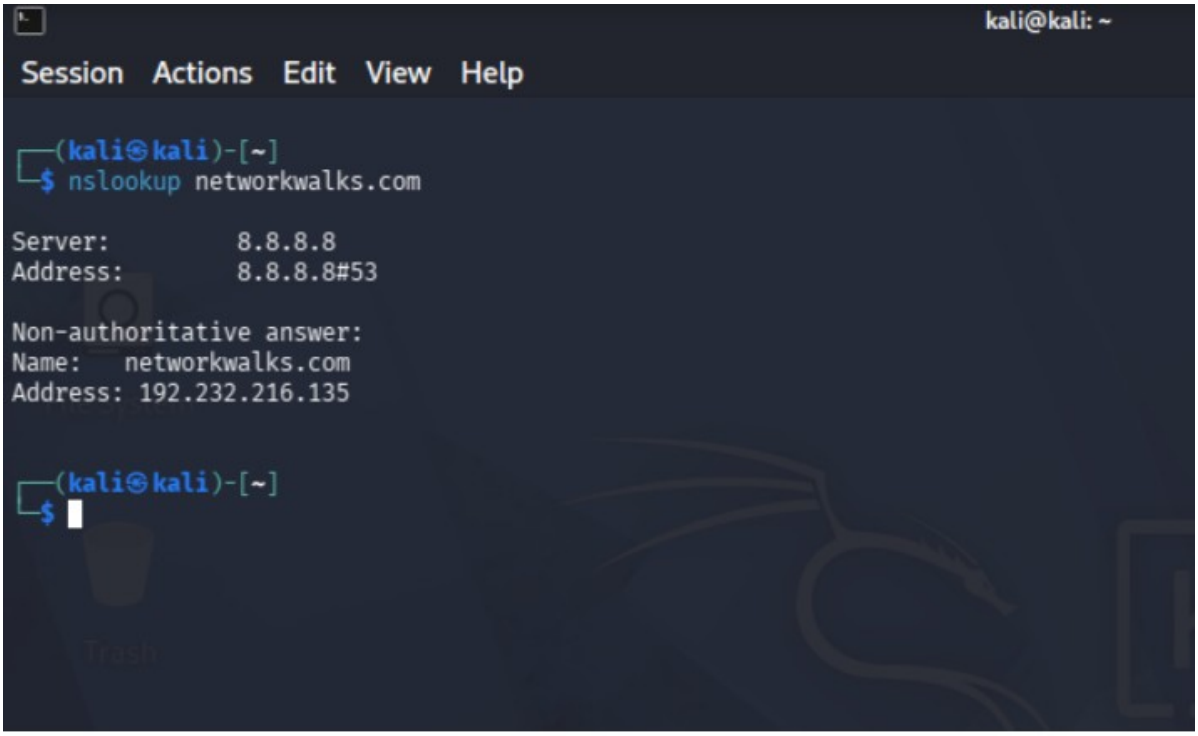
The screenshots show WHOIS registration output and WhatWeb technology fingerprinting output for networkwalks.com.

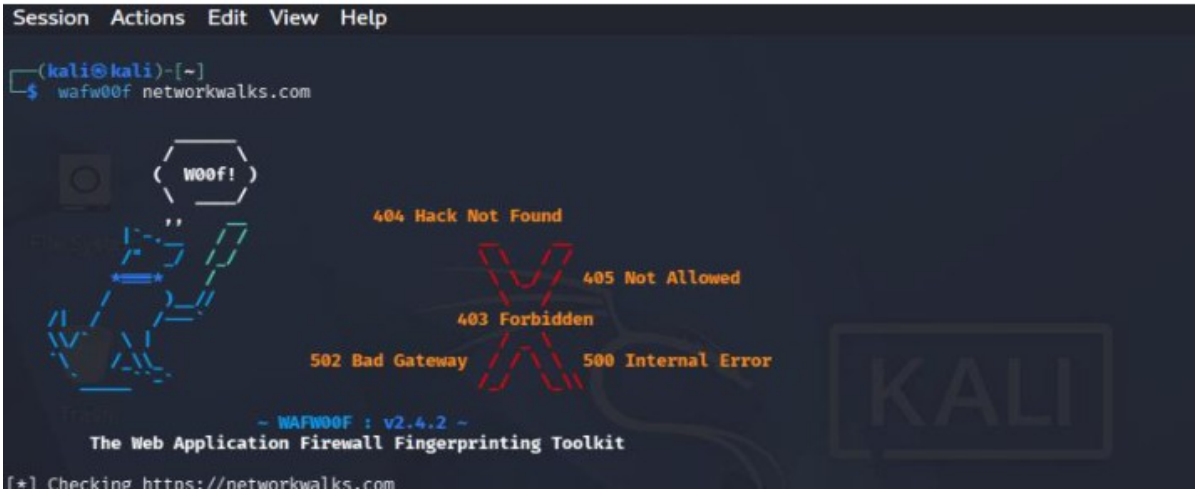




Evidence 2 – DNS Resolution and WAF Identification

The screenshots show nslookup resolving networkwalks.com to 192.232.216.135 and Wafw00f identifying ModSecurity (SpiderLabs).

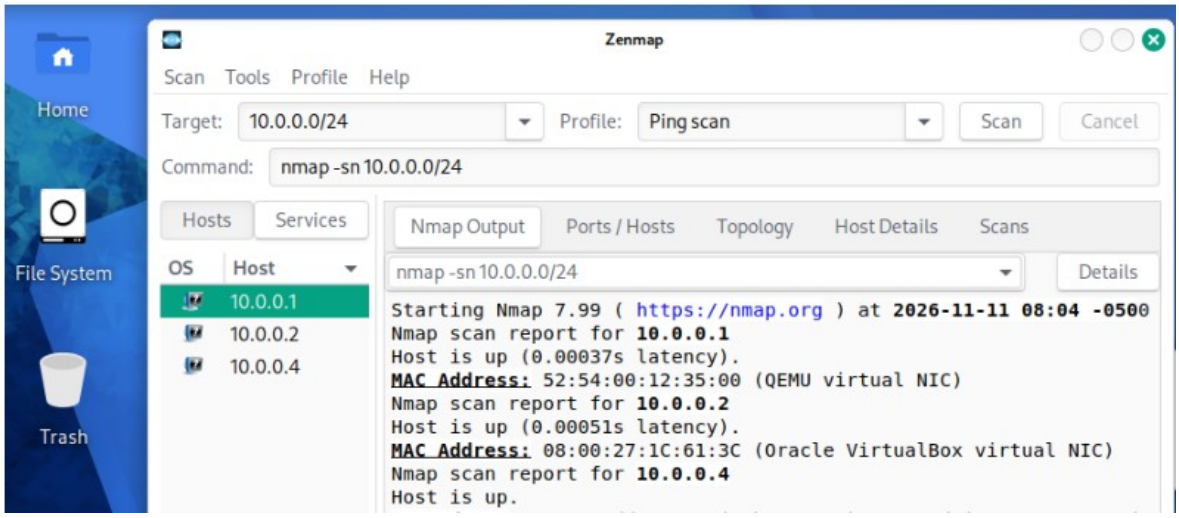




Evidence 3 – HTTP Headers and Zenmap Host Discovery

The screenshots show curl response headers and a Zenmap Ping Scan of 10.0.0.0/24. The visible Zenmap output lists three responding hosts.





10. Author and Project Information

FIELD	DETAIL
Author	Debela Beyena
Program	Cybersecurity Program at Networkwalks
Batch	B084
Week	02
Project	W2-PM-FINAL – Footprinting and Network Scanning
Repository	Debelabeyena2026/Footprinting-Reconnaissance-Attacks-with-Multiple-Kali-Tools

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